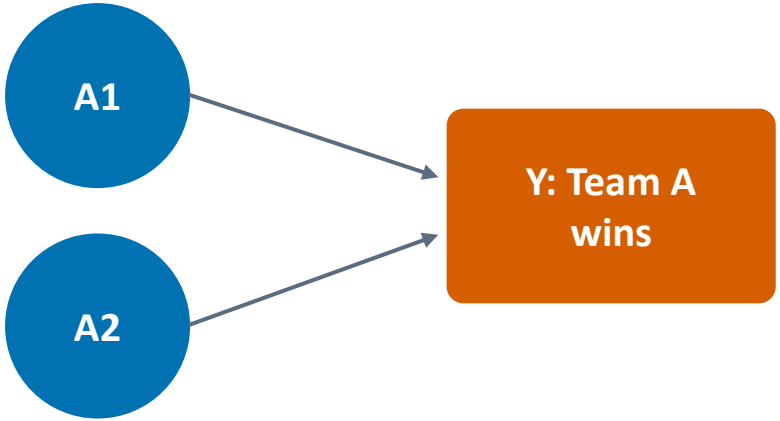
Individual-Level Evidence for Temporal Bayesian Player-Skill Estimation

A controlled ablation | CSE 516 Probabilistic Graphical Models | Group 3

Dhruvi Shah, Divija Nayak, Veera Bharakhada

M4 final presentation

The problem: a team result is shared



One outcome, two contributors

- A win credits both teammates equally
- Frequent partners are statistically hard to tell apart
- Team-only models identify strength of pairs, not individuals
- Extra per-player evidence may disambiguate them

Research question and scope

Hypothesis: adding individual observations Z to shared outcomes Y lowers the error of recovered skills, under otherwise identical conditions.

12	30	10	70:30
players	time points	paired seeds (42–51)	chronological split

Synthetic data with known truth. Not real ATP validation, not a full TrueSkill Through Time reproduction.

Why a probabilistic graphical model

1 Latent skills

Skills are never observed directly, so they are hidden nodes.

2 Temporal dynamics

A Gaussian random walk lets evidence flow between time points.

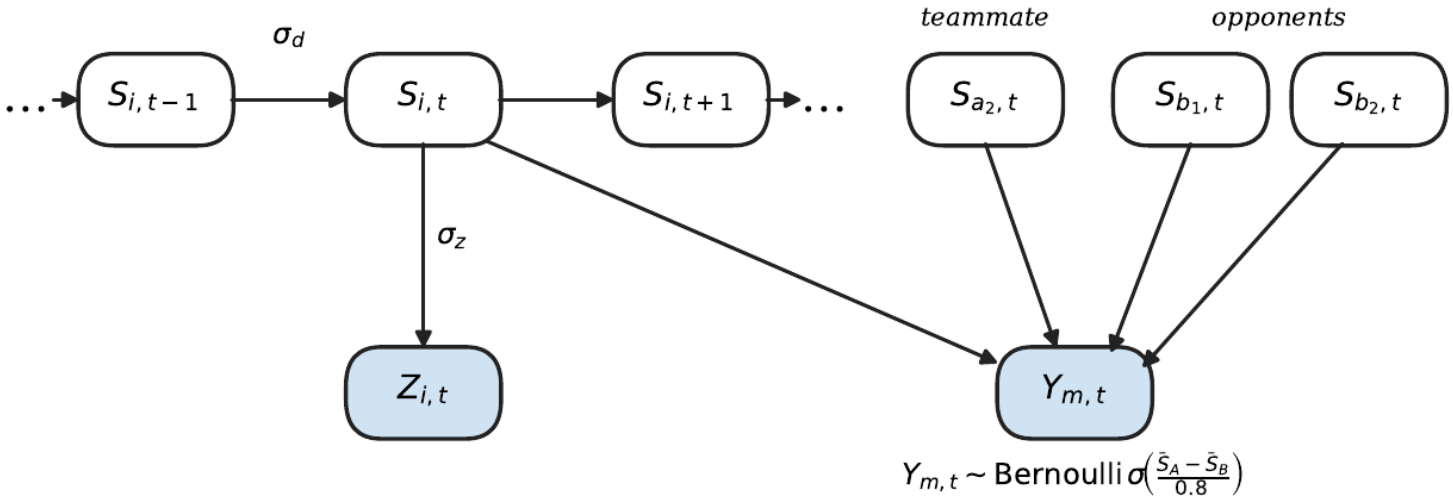
3 Shared evidence

Outcome nodes have both teammates as parents, which couples them.

4 Auditable assumptions

Conditional independencies are explicit. A new evidence source is one added factor.

Graphical representation



Shaded = observed (Z only in the extension; Y in both models). S: latent skill, Z: individual observation, Y: shared team outcome. Player i is shown as the focal member a_1 of team A; the plate repeats over players i, times t and matches m at each t.

Baseline = same graph without Z. Shaded = observed.

Generative model and factorisation

Skill prior and drift

$S_{i,0} \sim N(0, 1^2)$ $S_{i,t} \mid S_{i,t-1} \sim N(S_{i,t-1}, 0.12^2)$

Team outcome

$Y \sim \text{Bernoulli}(\sigma((S_a - S_b) / 0.8))$

Individual observation (extension only)

$Z \sim N(S_{i,t}, 0.65^2)$

Independence assumptions

- Skills are Markov in time
- Given the four players' skills, a match is independent of other matches
- Given its own skill, Z is independent of Y and other Z

Learning and inference

1 Learning: MAP

- Minimise negative log joint: prior + logistic outcome + (extension) Gaussian observation terms
- Convex; SciPy L-BFGS-B, analytic gradients
- Hyperparameters fixed, no structure learning

2 Inference: Laplace

- Covariance = inverse Hessian at the mode, posterior SD from its diagonal
- Held-out win probability uses fitted skills only, no test evidence
- $d = 360$ states; dense $O(d^3)$ inverse is cheap here, not at scale

Experimental design and baseline

Fit: t = 0–20 (252 matches)	Held out: t = 21–29

Baseline: team-only

- Same graph, prior, optimiser, split
- Sees only team outcomes Y

Extension: + individual

- Identical, plus Z factors
- Z = skill + noise (SD 0.65)
- Synthetic generator, seeds 42–51

KPIs and statistics

Skill recovery

MAE and RMSE vs. truth on held-out times

Probability quality

Log-loss (= NLL), Brier score, reliability diagram

Cost and uncertainty

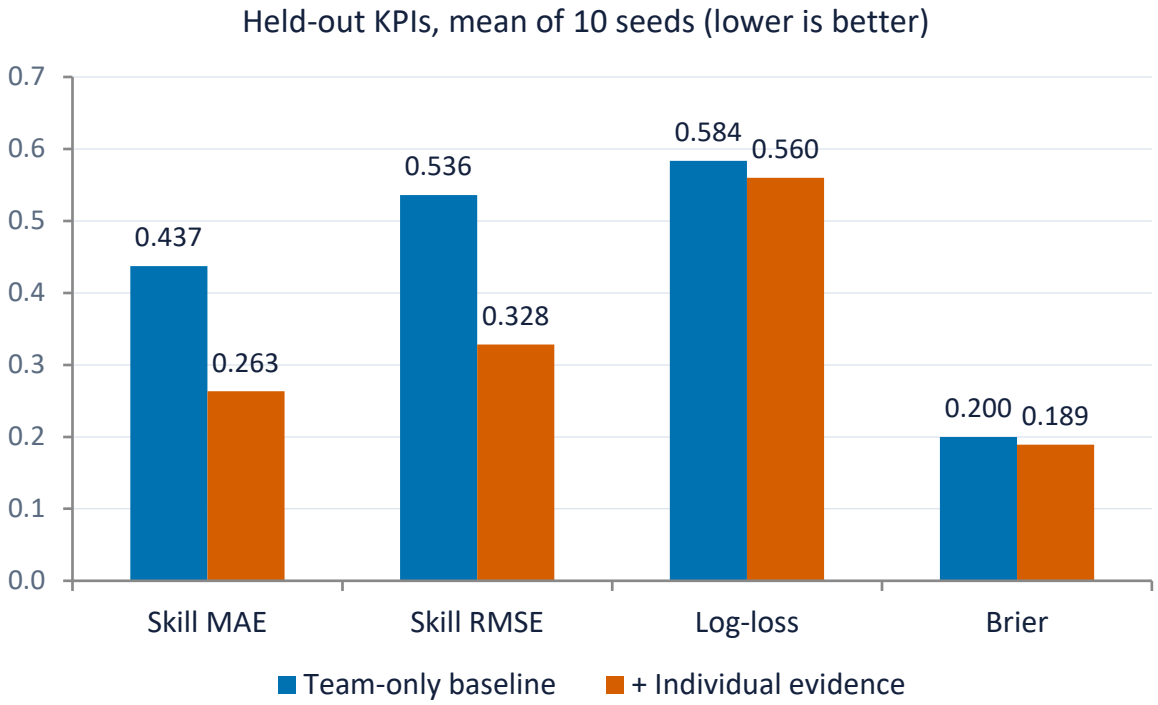
Runtime, L-BFGS-B iterations, posterior SD

Statistics

Mean ± SD, 95% CI, paired t, Wilcoxon, bootstrap, Holm, effect size d_z

Lower errors and computational costs are preferable. Posterior SD measures uncertainty, not necessarily performance.

Results: improved skill recovery and held-out prediction



−40%

skill MAE

0.437 → 0.263

10 / 10

seeds improve on MAE

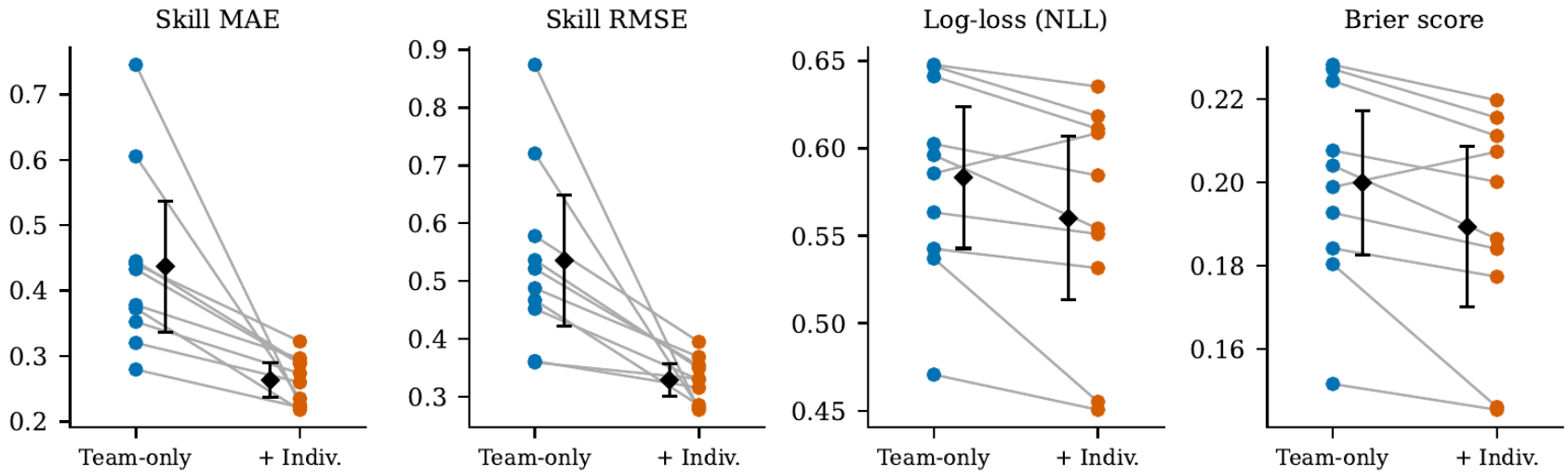
9 / 10 on NLL, Brier

2.4×

fewer iterations

165 → 67

Seed by seed: a consistent gain



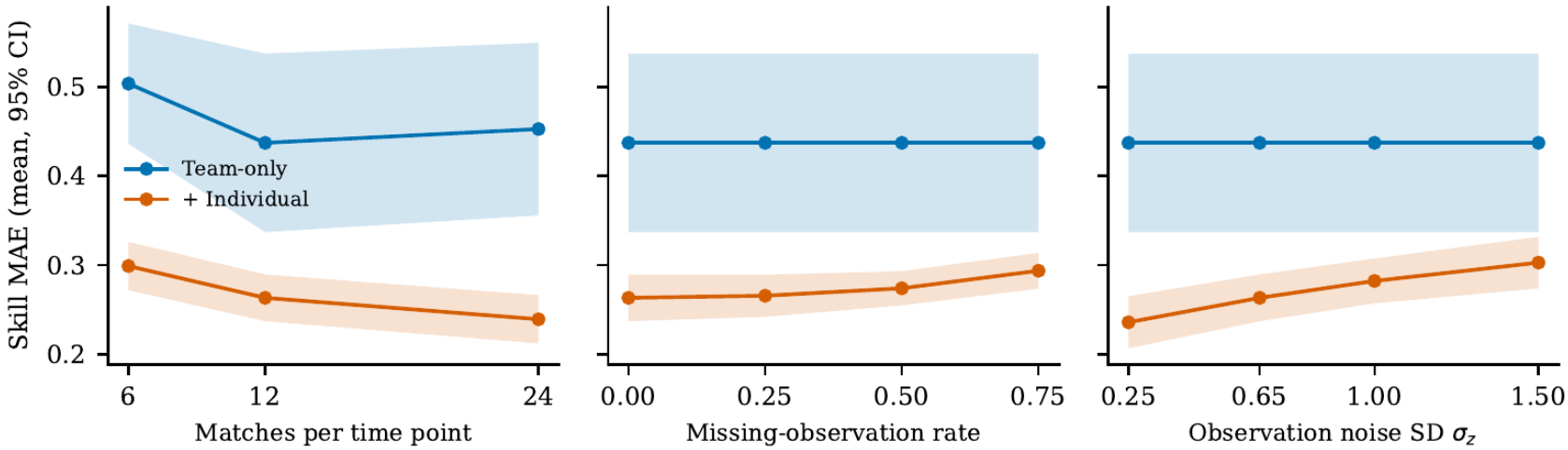
- Skill MAE and RMSE improve in 10/10 seeds; NLL and Brier in 9/10
- Baseline MAE SD 0.140 vs 0.037: without individual evidence recovery is also less stable (seed 47: 0.745 vs 0.226)

Statistical analysis

KPI	Δ (base − ext)	95% t-CI	p (paired t)	p (Holm)	d_z
Skill MAE	0.1740	[0.066, 0.282]	0.0055	0.020	1.15
Skill RMSE	0.2073	[0.081, 0.334]	0.0049	0.020	1.17
Log-loss (NLL)	0.0234	[0.004, 0.043]	0.022	0.024	0.87
Brier	0.0106	[0.003, 0.018]	0.012	0.024	0.99

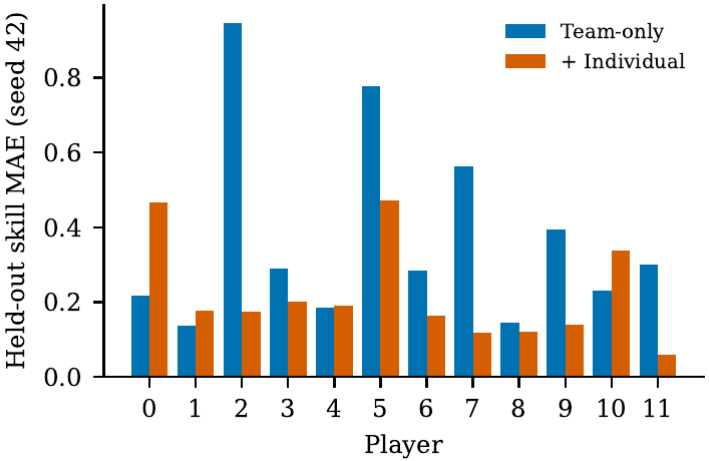
- MAE: bootstrap 95% CI [0.097, 0.274]; Wilcoxon p = 0.002
- Gain is large for skill recovery, small for outcome prediction
- Intervals reflect variation across 10 simulated seeds only

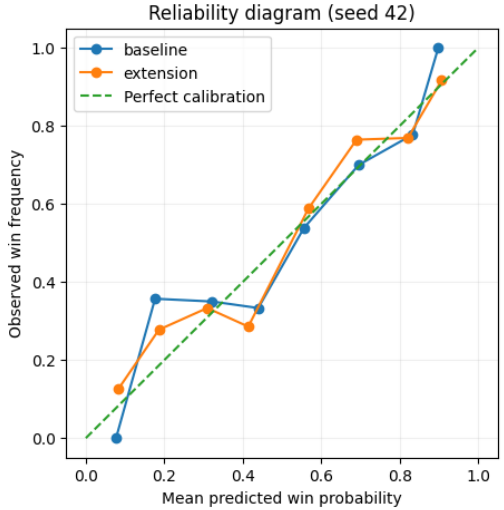
Robustness: the gain persists but narrows



- Missing 0 → 75%: MAE gain 0.174 → 0.144
- Noise σ 0.25 → 1.5: gain 0.201 → 0.134
- 6 matches/time: skill gain significant ($p < 0.001$), NLL gain not ($p = 0.25$)

Error analysis and calibration





8 of 12

players improve in seed 42; players 0, 1, 4, 10 get worse

- Not uniformly better: single-seed view, no population claim
- Reliability diagram is descriptive only; ECE was not computed

Positioning against the SOTA anchor

Identify	Anchor: TrueSkill Through Time package (Landfried & Mocskos, 2025)
Understand	Temporal skill graph, evidence propagated by message passing
Compare	Ours: logistic team likelihood, MAP + Laplace, 12 synthetic players
Validate	No head-to-head run. Only the internal ablation is validated
Reposition	M1 broad TTT study → narrow proof-of-concept. No superiority claim

Limitations and next step

What this does not show

- Evidence is simulated from truth, so informative by construction
- 12 synthetic players, one generator, 10 seeds
- Simplified model, not full TTT
- Model noise matches generator: no mis-specification test

Next research step

1. Real individual-level doubles features
2. Empirical observation likelihood with bias and missingness
3. Learn σ_d, τ, σ_z instead of fixing them
4. Calibration error, then benchmark vs. published TTT

What we learned

- Individual evidence improves teammate skill estimation: skill MAE −40%, 10/10 seeds, Holm $p = 0.020$
- Predictive gains are smaller: NLL −0.023, Brier −0.011
- Benefit shrinks with noise and missingness; not uniform across players
- A proof-of-concept relative to the TTT anchor, not a real-data result